# Multimedia Signal processing

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1. **Question (8 pts)**:
   Consider a continuous source $X$, whose p.d.f. is given by

$$p_X(x) = \frac{1}{2}e^{-|x|} \tag{1}$$

   (a) Design the Lloyd-Max quantizer for the source, using $M = 4$ quantization bins. Initialize the algorithm using the following thresholds $\{-\infty, -1, 0, 1, +\infty\}$. Stop after the first *complete* iteration.

   (b) Compute the entropy, in bits / symbol, of the source representing the output of the Lloyd-Max quantizer.

   Hint. You might find the following indefinite integral useful

$$\int ae^{ca}da = \frac{e^{ca}}{c^2}(ca-1) \tag{2}$$

   **Solution**

   (a) The pdf is symmetric. Hence, the Lloyd-Max quantizer will also be symmetric. Hence, we focus on the quantization regions $I_3$ and $I_4$ and the corresponding reconstruction levels $y_3$ and $y_4$.

   The Lloyd-Max quantizer is obtained by iterating the following steps:

   - Given the quantization regions, find the optimal reconstruction levels as the centroid of the corresponding region. That is,

$$y_k = \frac{\int_{a\in I_k} ap_X(a)da}{\int_{a\in I_k} p_X(a)da} \tag{3}$$

   Hence, we need to compute

$$\int_{a\in I_3} ap_X(a)da = \int_0^1 ae^{-a}da = \frac{1}{2}\left[e^{-a}(-a-1)\right]_0^1 = \frac{1}{2}(1-2e^{-1}) \tag{4}$$

$$\int_{a\in I_4} ap_X(a)da = \int_1^{+\infty} ae^{-a}da = \frac{1}{2}\left[e^{-a}(-a-1)\right]_1^{+\infty} = e^{-1} \tag{5}$$

   The reconstruction levels are

$$y_3 = \frac{1-2e^{-1}}{1-e^{-1}} \tag{6}$$

$$y_4 = 2 \tag{7}$$

- Given the reconstruction levels, the thresholds of the quantization regions are obtained as the midpoints of the reconstruction levels. That is,

$$a_1 = -\infty \tag{8}$$
$$a_2 = -a_4 \tag{9}$$
$$a_3 = 0 \tag{10}$$
$$a_4 = \frac{y_3 + y_4}{2} = \frac{3}{2(1 - e^{-1})} \tag{11}$$
$$a_5 = +\infty \tag{12}$$

  Then, the would proceed iterating these two steps.

(b) The probabilities of the output symbols of the quantizer can be found as follows

$$p_k = \int_{a \in I_k} p_X(a) da \tag{13}$$

Hence:

$$p_3 = \int_0^{a_4} \frac{e^{-a}}{2} da = \frac{1}{2} \left[ -e^{-a} \right]_0^{a_4} = \frac{1}{2} \left[ 1 - e^{-a_4} \right] \tag{14}$$
$$p_4 = \int_{a_4}^{+\infty} \frac{e^{-a}}{2} da = \frac{1}{2} \left[ -e^{-a} \right]_{a_4}^{+\infty} = \frac{1}{2} e^{-a_4} = \dots \tag{15}$$

By symmetry, $p_1 = p_4$ and $p_2 = p_3$.

The value of the entropy can be computed as follows

$$H(\hat{X}) = -\sum_{k=1}^{4} p_k \log_2 p_k \tag{16}$$

2. **Question (5 pts)**
   (a) Illustrate the design of analysis-by-synthesis speech coders.
   (b) Discuss the use of long- and short-term predictors.

   **Solution**

   See slide set `MMSP_2.1_Speech coding`.

3. **Question (5 pts)**
   (a) Illustrate the rate-distortion curve for discrete and continuous sources, specifying the achievable region. Express the distortion in terms of quantization SNR.
   (b) Compare different scalar quantization strategies in terms of their operational rate-distortion curves.

   **Solution**

   See slide set `MMSP_1.2_Lossy_Coding`.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**

   (a) Consider a discrete source

   - [T] [F] $H_N(x) \geq H_{N+1}(x)$, where $H_N(x) = \frac{H(x_1,\ldots,x_N)}{N}$
     TRUE. Increasing the block size reduces the lower bound on the number of bits needed to encode a source.
   - [T] [F] $H_\infty(x) = H(x)$ iff the source is memoryless.
     TRUE. It the source has memory, $H_\infty(x) < H(x)$
   - [T] [F] The average code length of an Huffman code $L_H$ is such that $H(x) \leq L_H < H(x) + 1$
     TRUE.
   - [T] [F] In a Huffman code, the two longest codewords have always the same length.
     TRUE. This follows directly from the construction of the Huffman tree.

   (b) Consider the KLT.

   - [T] [F] The basis functions are orthonormal.
     TRUE. The KLT is obtained by taking the eigenvectors of the autocorrelation matrix, which are orthonormal.
   - [T] [F] The autocorrelation matrix of the transform coefficients is diagonal.
     TRUE. By construction, the KLT produces transform coefficients which are uncorrelated. Hence, the autocorrelation matrix is diagonal.
   - [T] [F] Given the KLT learned on an image collection, the coding gain of the KLT is higher than that of the DCT, for each image in the collection.
     FALSE. The KLT captures the statistical properties of the whole collection. On a single image, it might be possible that the DCT leads to a larger coding gain than the DCT.

   (c) Consider an image coding system

   - [T] [F] Spatial redundancy is mostly exploited by means of DPCM.
     FALSE. Although DPCM might be used to encode the residual spatial redundancy of DC coefficients.
   - [T] [F] Threshold quantization leads to higher coding efficiency than zonal quantization.
     TRUE.
   - [T] [F] In JPEG, the chroma components are subsampled by a factor of 2 both horizontally and vertically.
     TRUE. This is done to exploit the fact that the HVS is more sensitive to changes in luminance than in color.

5. **Question (Matlab - 10pts)**

   Consider the image *image.png*. Arrange the pixel values of the image in two vectors, $\mathbf{x}_r$ and $\mathbf{x}_c$, obtained by reading the image in raster scan order either row-wise or column-wise.

   (a) Design a DPCM coder to lossy code both $\mathbf{x}_r$ and $\mathbf{x}_c$. Use a scalar uniform quantizer, whose step size $\Delta$ can be adjusted, ignoring the saturation region.

   (b) By varying $\Delta$, compute the operational rate-distortion curves for both $\mathbf{x}_r$ or $\mathbf{x}_c$. Estimate the rate as the entropy of source representing the output symbols of the quantizer.

   (c) Plot the rate-distortion curves (PSNR vs. rate) and comment the result.

   **Solution**

```
%% 2013_03_01_MMSP_exam

clear all
close all
```

```
clc

%% 1) Consider the image 'image.png'. Arrange the pixel values in two vectors
%%    xr (row-wise reading) and xc (column-wise reading)
Image = double(imread('image.png'));
aux = Image';
xr = aux(:);
xc = Image(:);

%% 2) Design a DPCM coder with adjustable step size
R = 1:8;
M = 2.^R; % number of levels
rate_r = zeros(length(M),1);
psnr_r = zeros(length(M),1);
rate_c = zeros(length(M),1);
psnr_c = zeros(length(M),1);

for m = 1:length(M)
    Q = (255 - (-255))/M(m);

    %% row-wise reading
    dr_tilde = zeros(length(xr),1);
    xr_hat = zeros(length(xr),1);

    % first sample
    dr_tilde(1) = Q*round(xr(1)/Q);
    xr_tilde = dr_tilde(1);
    xr_hat(1) = xr_tilde;

    % successive samples
    for i = 2:length(xr)
        d = xr(i) - xr_tilde;
        dr_tilde(i) = Q*round(d/Q);
        xr_tilde = dr_tilde(i) + xr_tilde;
        xr_hat(i) = dr_tilde(i) + xr_hat(i-1);
    end

    % estimate the entropy
    h = hist(dr_tilde(:),-255:255);
    p = h / sum(h);
    p = p(p~=0);
    rate_r(m) = -sum(p.*log2(p));

    % estimate the psnr
    E = xr_hat - xr;
    MSE = sum(sum(E.^2))/(length(E));
    psnr_r(m) = 10*log10(255^2/MSE);

    %% column-wise reading
    dc_tilde = zeros(length(xc),1);
    xc_hat = zeros(length(xc),1);

    % first sample
    dc_tilde(1) = Q*round(xc(1)/Q);
    xc_tilde = dc_tilde(1);
    xc_hat(1) = xc_tilde;

    % successive samples
    for i = 2:length(xc)
        d = xc(i) - xc_tilde;
        dc_tilde(i) = Q*round(d/Q);
        xc_tilde = dc_tilde(i) + xc_tilde;
        xc_hat(i) = dc_tilde(i) + xc_hat(i-1);
    end

    % estimate the entropy
    h = hist(dc_tilde(:),-255:255);
    p = h / sum(h);
    p = p(p~=0);
    rate_c(m) = -sum(p.*log2(p));

    % estimate the psnr
    E = xc_hat - xc;
    MSE = sum(sum(E.^2))/(length(E));
    psnr_c(m) = 10*log10(255^2/MSE);
end

%% Plot PSNR vs rate
figure(1)
plot(rate_r,psnr_r,'b'),hold on, grid on, plot(rate_c,psnr_c,'r'),hold off
legend('row-wise reading','column-wise reading','Location','BestOutside')
ylabel('PSNR [dB]'), xlabel('Rate')
```